

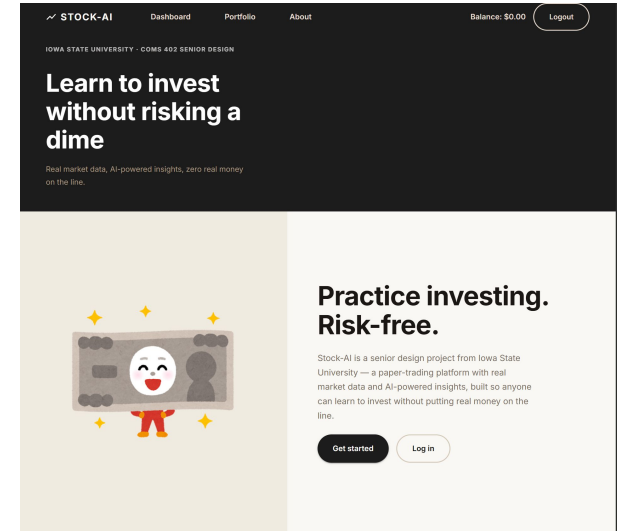
Stock-AI Demo #2

Team UG_HF_03

Andrew Butler, Matthew He, Ethan Roe, Aaron Trelstad

What is Stock-AI?

- Learning stock trading is difficult
- Trading & trading platforms are expensive
- Create Free learning platform
 - Mock trades, so no risk
 - AI assistance (ask questions, provide recommendations)



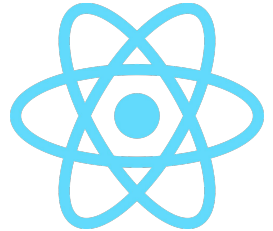
Technologies in use

Backend

- MariaDB
- FastAPI Python framework
- Podman for containerized deployment

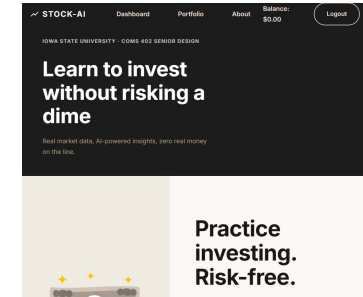
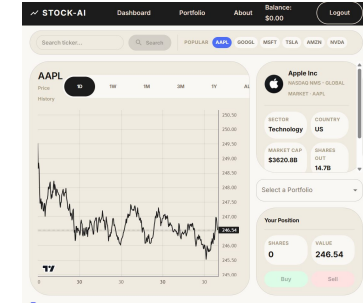
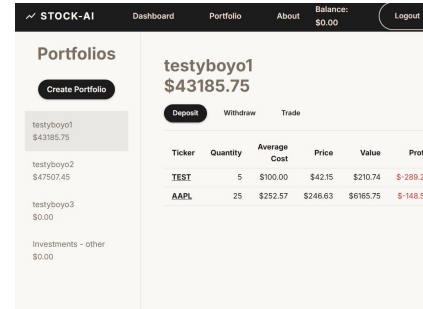
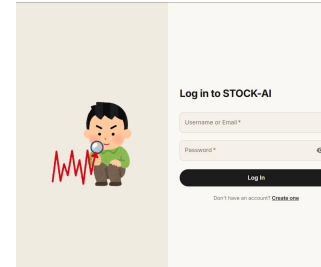
Frontend

- React.js web framework
- Material UI component library



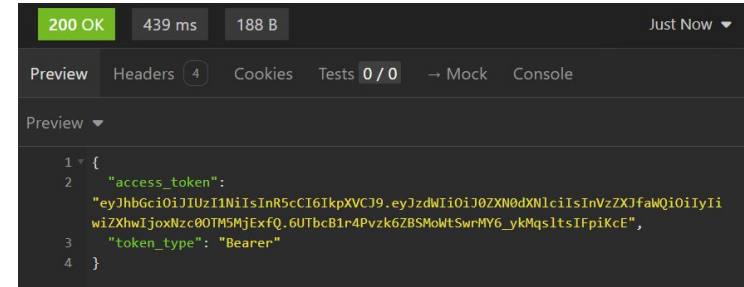
Stock-AI Functionality

- Typical website design
 - Login, register page
 - Dashboard page
 - Portfolios page
 - Leagues page
 - About page



Implementation - Login & Security

- Password hashing
- Login generates Json Web Token (JWT)
 - Gives token
 - Contains user ID, username
 - Validates future action
 - Expires after 60 minutes
- Protected routes
 - Anything that is user specific requires valid JWT
 - Endpoints check JWT validity
 - Endpoints check user ownership of resource (portfolio, conversation, etc)
- Suboptimal: Implemented manually



A screenshot of a REST client interface showing a successful HTTP response. The status bar at the top indicates a 200 OK response with a response time of 439 ms and a body size of 188 B. The response body is a JSON object containing an access token and its type.

```
1 {  
2   "access_token":  
   "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOiI0ZDZlN0dXNlciIsInVzZXJfaWQiOiIyIiwiaXNzZXIjOiJ0OTM5MjExfQ.6UTbcB1r4Pvzk6ZBSMoWtSwrMY6_ykMqs1tsIFpiKcE",  
3   "token_type": "Bearer"  
4 }
```

Implementation - Trading

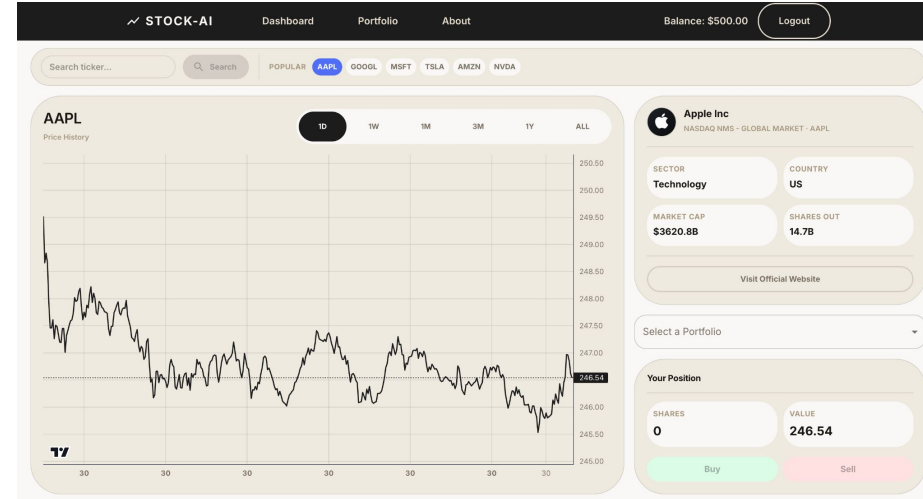
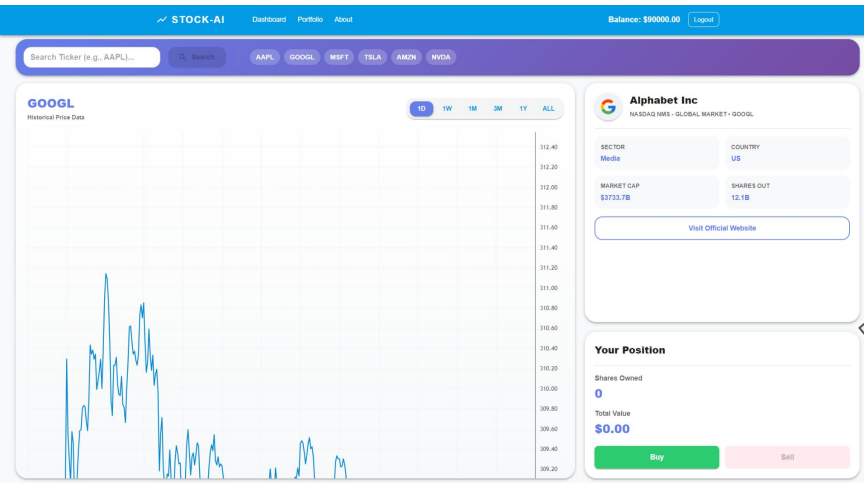
- Users can have multiple portfolios
- User & Portfolio cash balances are handled separately
 - User can transfer money in & out of a given portfolio
- What data we store:
 - User & portfolio cash balance
 - Held positions
 - Trade history (eventually viewable)
 - Average cost basis for position
 - Ticker symbol (e.g., AAPL)

	id	symbol	name
▶	1	TEST	TestCo
	3	AAPL	Apple Inc
	4	GOOGL	GOOGL
	5	MSFT	Microsoft Corp
	9	NVDA	NVIDIA Corp
	10	TSLA	Tesla Inc
	11	AMZN	Amazon.com Inc
•	NULL	NULL	NULL

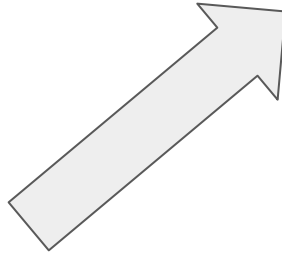
	id	portfolio_id	ticker_id	quantity	avg_cost_basis
▶	6	1	1	5.000000	100.0000
	7	1	3	25.000000	252.5700
	8	5	4	2.000000	0.0000
	12	6	4	1.000000	274.2300
	13	2	9	15.000000	166.1700
	14	4	4	10.000000	274.1700
	15	4	10	20.000000	357.1700
	16	4	9	10.000000	166.2500
	17	17	3	1.000000	246.5400
	19	29	3	5.000000	246.5400
	20	30	5	2.000000	358.7600
	21	29	4	2.000000	273.5100
	22	27	3	5.000000	246.5400
	23	27	4	3.000000	273.5100
	24	27	9	5.000000	165.0600
	25	6	11	3.000000	200.9000
	26	32	3	4.000000	246.5400
•	NULL	NULL	NULL	NULL	NULL

	id	portfolio_id	ticker_id	trade_type	quantity	price	realized_pnl	executed_at
▶	1	4	1	BUY	10.500000	250.0000	NULL	2026-02-24 21:13:02
	2	4	1	BUY	5.000000	250.0000	NULL	2026-02-24 21:24:03
	3	4	1	BUY	5.000000	150.0000	NULL	2026-02-24 21:24:14
	4	4	1	BUY	0.500000	0.0000	NULL	2026-02-24 21:24:39
	5	4	1	BUY	4.000000	-10.0000	NULL	2026-02-24 21:24:54
	6	4	1	BUY	5.000000	-10.0000	NULL	2026-02-24 21:27:11
	7	4	1	BUY	5.000000	0.0000	NULL	2026-02-24 21:34:27
	8	4	1	BUY	0.000000	0.0000	NULL	2026-02-24 21:34:33
	9	4	1	BUY	0.000000	0.0000	NULL	2026-02-24 21:34:36
	10	4	1	BUY	0.000000	0.0000	NULL	2026-02-24 21:34:44
	11	4	1	BUY	100.000000	250.0000	NULL	2026-02-24 21:35:21
	12	4	1	BUY	75.000000	250.0000	NULL	2026-02-24 21:36:29

Dashboard: UI Updates



Dashboard: UI Updates



 AAPL News

1

Yahoo

Just now

Western Digital, Photronics, Nova, FormFactor, and Amkor Stocks Trade Down, What You Need To Know

>

2

Yahoo

Just now

Amtech, Magnachip, Monolithic Power Systems, Allegro MicroSystems, and Kulicke and Soffa Shares Are Falling, What You Need To Know

>

3

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Just now

Himax, Teradyne, Entegris, IPG Photonics, and Universal Display Shares Plummet, What You Need To Know

>

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Just now

Seagate, Applied Materials, KLA Corporation, Lam Research, and MACOM Stocks Trade Down, What You Need To Know

>

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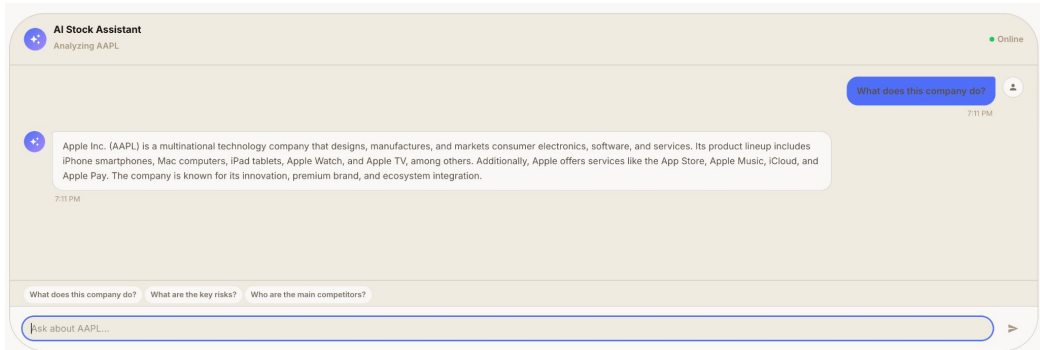
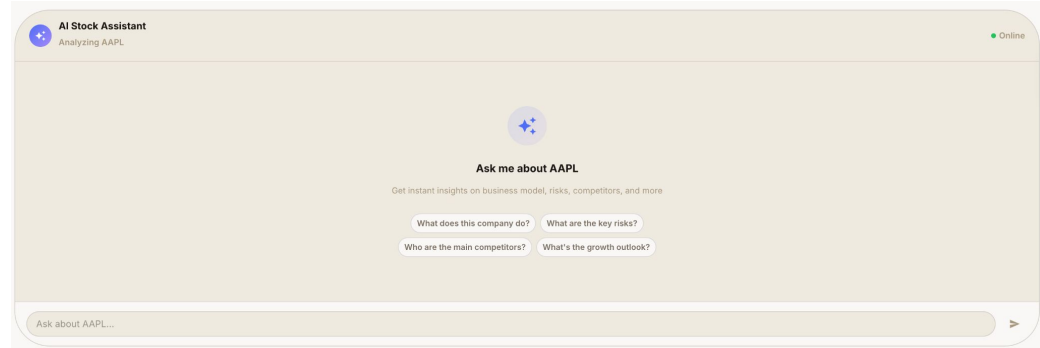
App Store

The I

Tim C

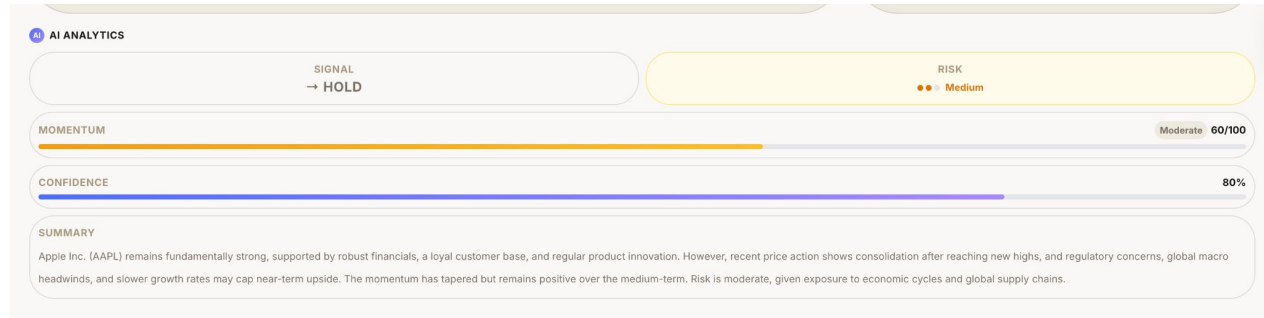
AI Chatbot

- OpenAI Conversation API

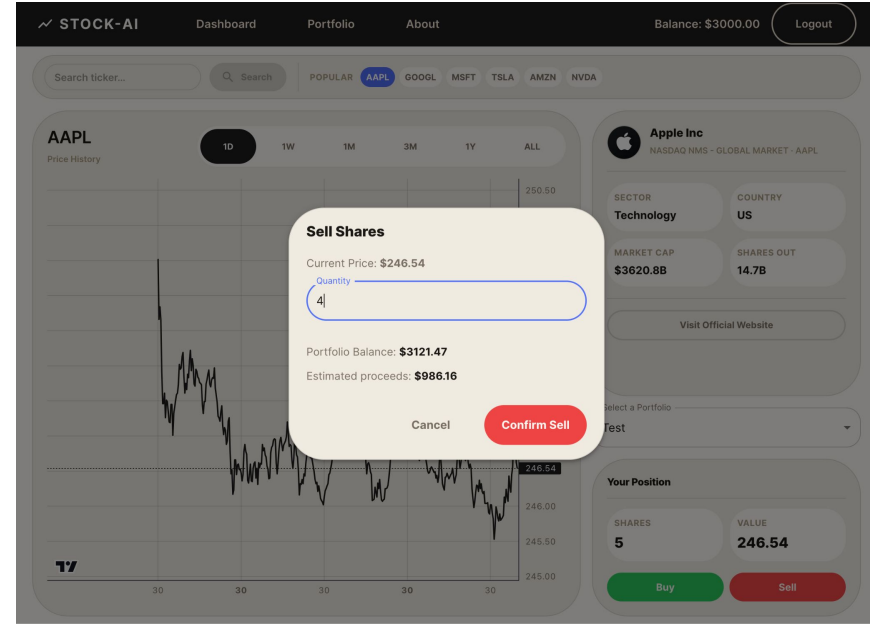
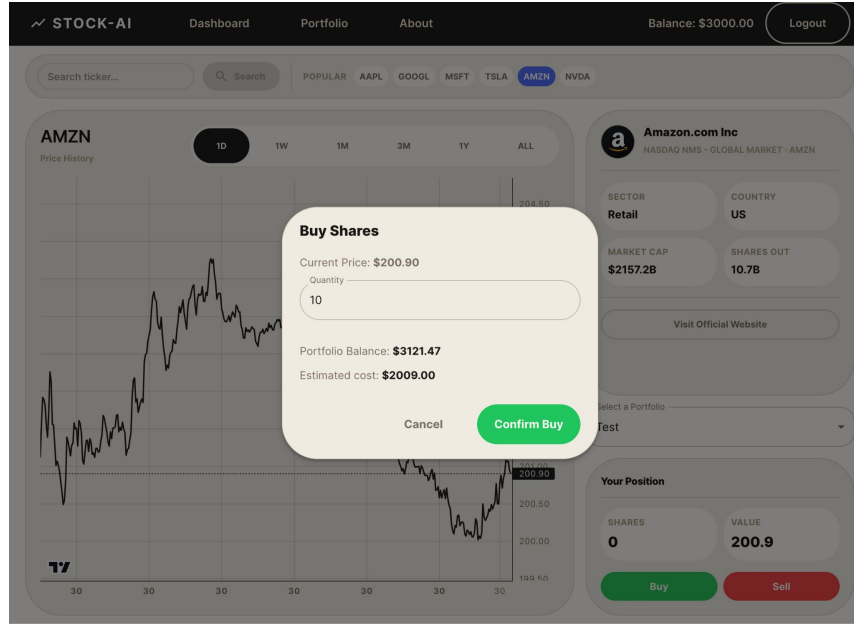


AI Analytics

- OpenAI Agents API

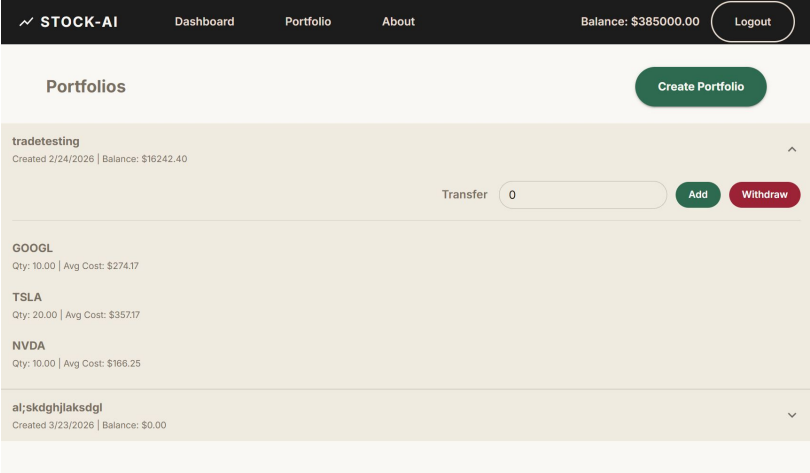


Mock Trading

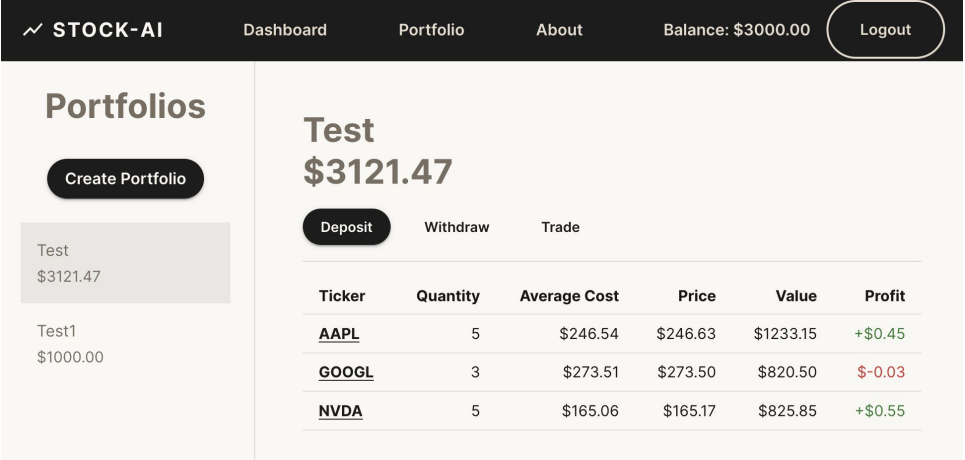


Portfolio UI/UX Improvements

Before:

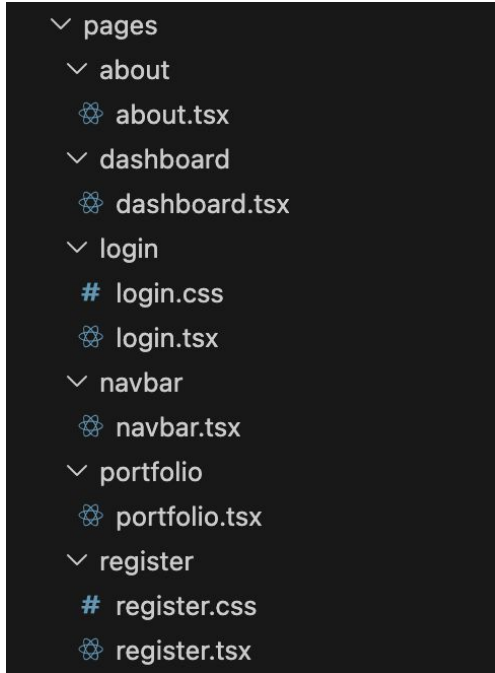


After:

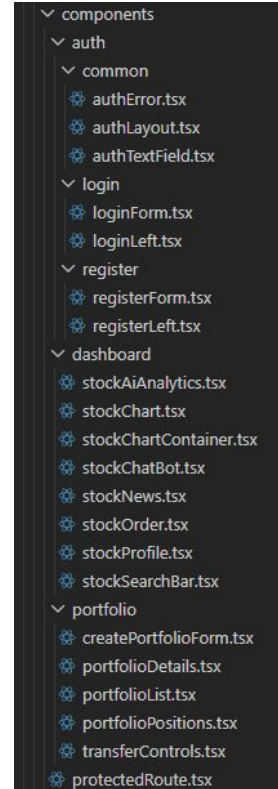


Frontend Code Refactor

Before:

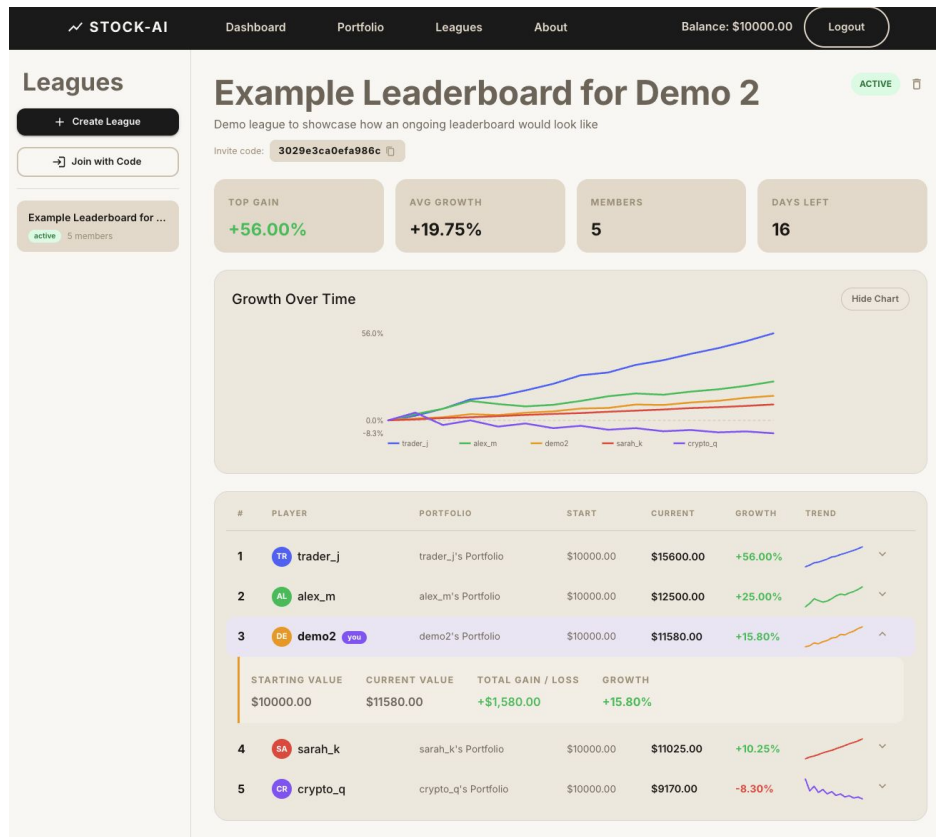


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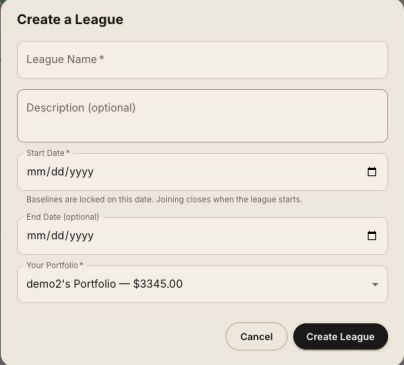


Competitive Leaderboards

- Engaging Stats
- Growth Chart
- Fantasy Sports Inspired Table
 - Sparklines
 - Expandable Rows
- Status Tags



Creating/Joining/Deleting Leaderboards



Create a League

League Name *

Description (optional)

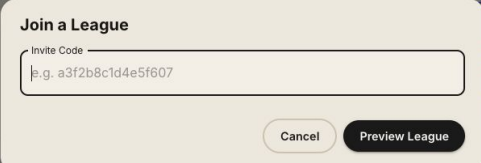
Start Date *
mm/dd/yyyy

Baselines are locked on this date. Joining closes when the league starts.

End Date (optional)
mm/dd/yyyy

Your Portfolio *
demo2's Portfolio — \$3345.00

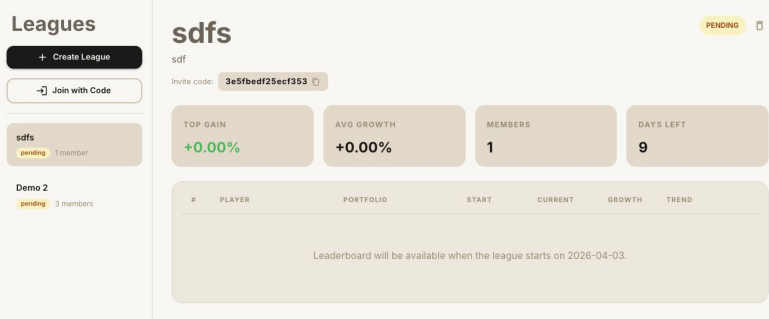
Cancel Create League



Join a League

Invite Code
e.g. a3f2b8c1d4e5f607

Cancel Preview League



Leagues

+ Create League

- Join With Code

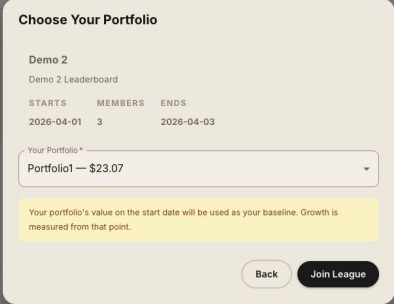
sdfs PENDING

Invite code: 3e5fbed25ecf353

TOP GAIN +0.00% AVG GROWTH +0.00% MEMBERS 1 DAYS LEFT 9

Demo 2 PENDING 3 members

#	PLAYER	PORTFOLIO	START	CURRENT	GROWTH	TREND
Leaderboard will be available when the league starts on 2026-04-03.						



Choose Your Portfolio

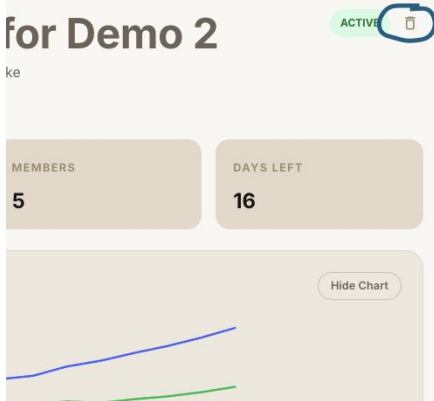
Demo 2
Demo 2 Leaderboard

STARTS	MEMBERS	ENDS
2026-04-01	3	2026-04-03

Your Portfolio *
Portfolio1 — \$23.07

Your portfolio's value on the start date will be used as your baseline. Growth is measured from that point.

Back Join League



for Demo 2 ACTIVE

MEMBERS 5 DAYS LEFT 16

Hide Chart

Competitive Leaderboard Architecture

- New Tables
 - League
 - League Member
 - Portfolio Snapshot
- Routes

leagues			^
POST	/leagues/create	Create a new league	🔒 ▼
GET	/leagues/mine	List leagues the current user belongs to	🔒 ▼
GET	/leagues/preview/{invite_code}	Preview a league by invite code (no auth required)	▼
GET	/leagues/{league_id}	Get league detail	🔒 ▼
DELETE	/leagues/{league_id}	Delete a league	🔒 ▼
POST	/leagues/{league_id}/join	Join a league with a chosen portfolio	🔒 ▼
GET	/leagues/{league_id}/leaderboard	Get ranked leaderboard for a league	🔒 ▼
POST	/leagues/{league_id}/snapshots	Trigger a snapshot for all members of a league	🔒 ▼
GET	/leagues/{league_id}/members/{member_id}/snapshots	Get snapshot time-series for a single member (race chart data)	🔒 ▼

UI redesign - Login/Register

Login

Username or Email *

Password *

Login Register



Log in to STOCK-AI

Username or Email *

Password * 

Log In

Don't have an account? [Create one](#)

Register

Username *

Email *

Password *

Register Back to Login

Create your account

We'll need your username, email address, and a unique password. You'll use this login to access STOCK-AI next time.

Username *

Email address *

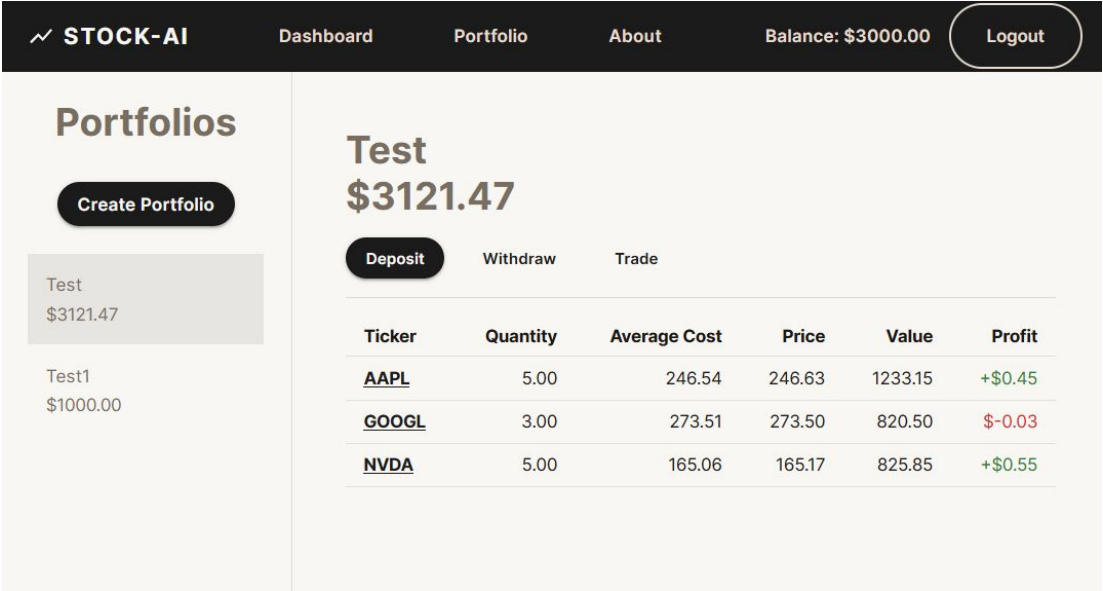
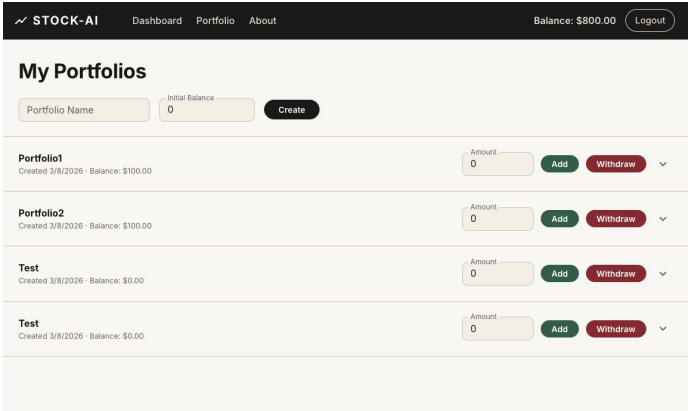
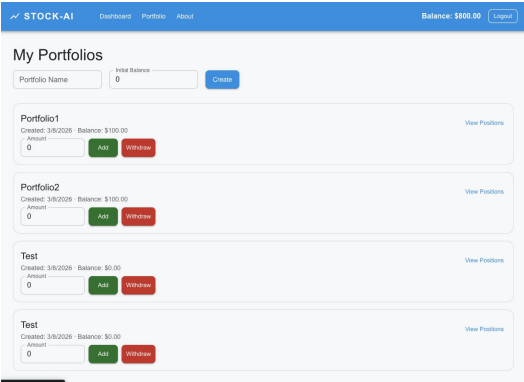
Password * 

Already have an account? [Log in](#)

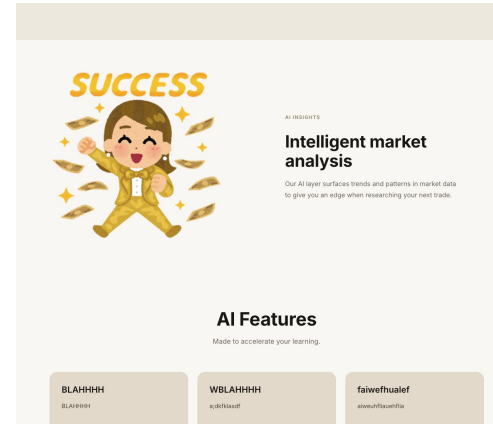
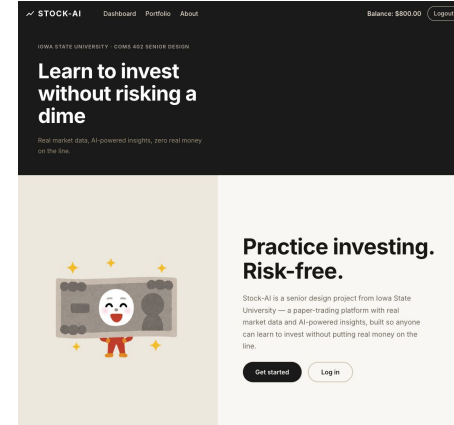
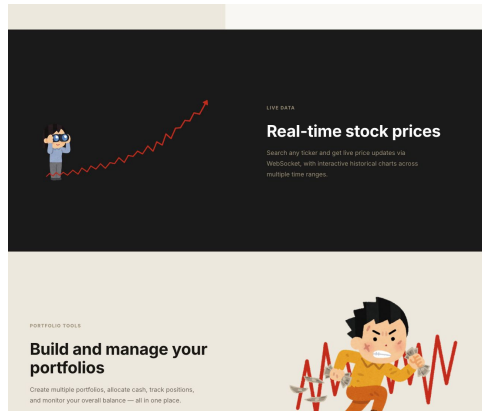
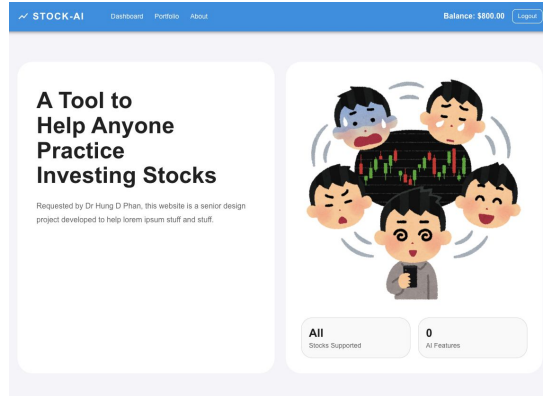
Continue



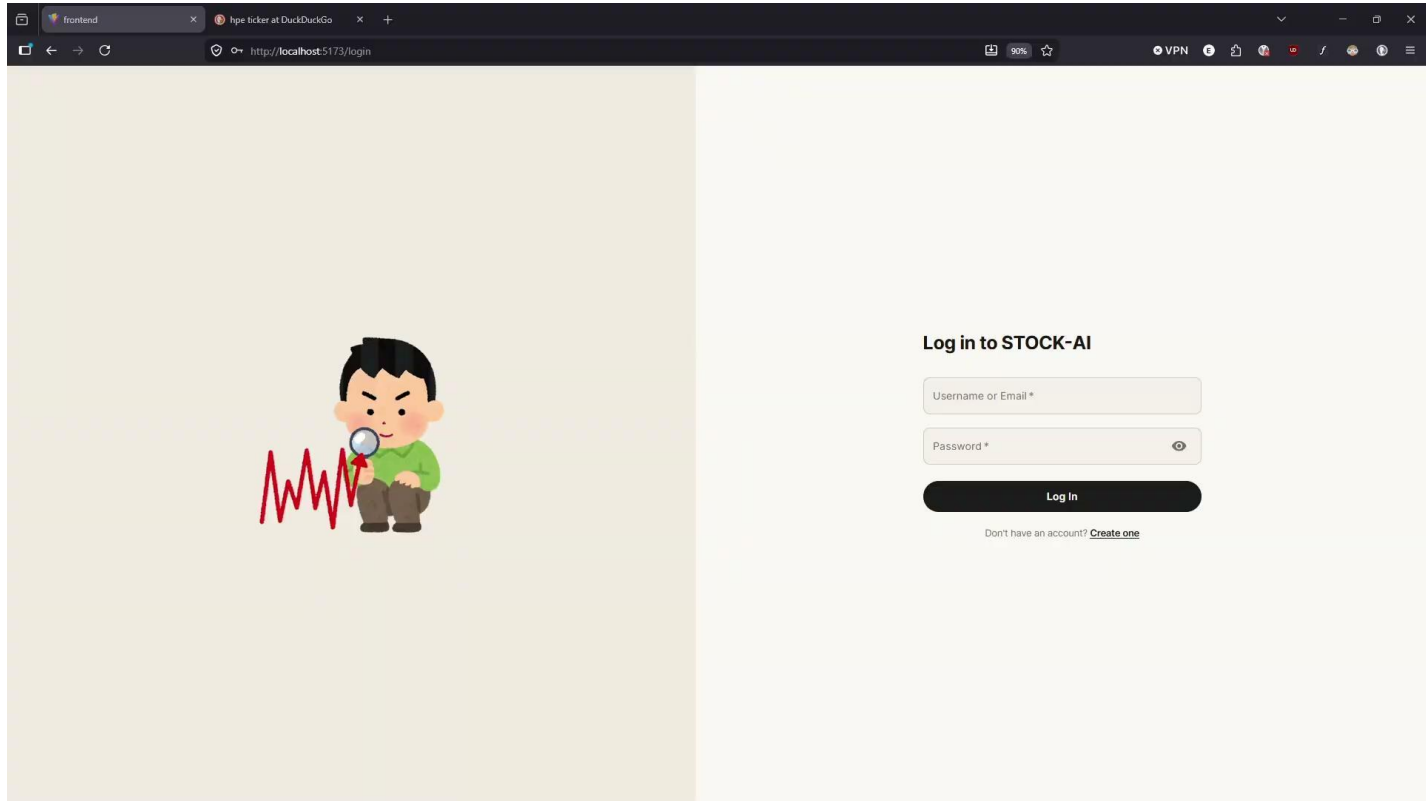
UI redesign - Portfolios



UI redesign - About Page



Demo



[Youtube upload if embed doesn't work](#)

Our goals moving forward

- Security refactor: FastAPI Users library
 - 'Forgot Password' pathway
 - Encrypt portfolio, trade data?
- Custom Watchlists
- Trade & chat history viewing
 - Endpoints already exist; just need UI
- AI model selection / graduate group model
- Further improvements to portfolio
 - Transferring trades between portfolios
 - Ability to delete or rename portfolios

Questions?